

# Prep Notes: Three Quick RSA Attacks

Håstad, Common Modulus, Cycling — built entirely from what you just taught

**Why these three, and in this order.** Each one reuses a tool you finished teaching in 1.1–1.2 an hour ago (CRT, Extended Euclidean Algorithm, Euler’s theorem), so there is nothing new to learn mathematically — only a new way of *using* it. Budget roughly 15 / 15 / 15 minutes. Suggested framing line for the class: “*Same theorem, opposite job: an hour ago it proved RSA correct, now it’s going to break it.*”

---

## 1. Håstad’s Broadcast Attack

(reuses: CRT)

**Setup.** Small public exponent (e.g.  $e = 3$ ), same message  $m$  broadcast to 3 recipients with independent, pairwise-coprime moduli  $n_1, n_2, n_3$ :

$$c_i = m^3 \bmod n_i, \quad i = 1, 2, 3.$$

**Core insight (say this first).** Three small, separately-opaque worlds (mod  $n_1$ , mod  $n_2$ , mod  $n_3$ ) get stitched by CRT into one big world (mod  $N = n_1 n_2 n_3$ ) — and against that much bigger modulus,  $m^3$  might not wrap around at all. It’s the “no modular wraparound” flaw from Lesson 1, resurfacing one level up.

**Theorem 1.** *If  $m < \min(n_1, n_2, n_3)$ , an eavesdropper who sees  $c_1, c_2, c_3$  recovers  $m$  — no private key needed.*

**Compressed proof.**

- CRT combines  $x \equiv c_i \pmod{n_i}$  ( $i = 1, 2, 3$ ) into a *unique*  $C \bmod N = n_1 n_2 n_3$ , computed directly from public data.
- $m^3$  also satisfies all three congruences (by definition of  $c_i$ )  $\Rightarrow$  by CRT’s uniqueness,  $C \equiv m^3 \pmod{N}$ .
- Since  $m < \min n_i$ , we get  $m^3 < N \Rightarrow C$  and  $m^3$  are both nonnegative integers  $< N$  that agree mod  $N$ , so they’re *exactly equal* as integers (not just congruent).
- Take an ordinary integer cube root of  $C$ . Done.

**Worked example (verified).**  $n_1 = 143 = 11 \cdot 13$ ,  $n_2 = 323 = 17 \cdot 19$ ,  $n_3 = 667 = 23 \cdot 29$  (pairwise coprime),  $m = 17$ :

$$c_1 = 17^3 \bmod 143 = 51, \quad c_2 = 17^3 \bmod 323 = 68, \quad c_3 = 17^3 \bmod 667 = 244.$$

$N = n_1 n_2 n_3 = 30,808,063$ . CRT-combining  $(51, 68, 244)$  gives  $C = 4913$ . Check:  $17^3 = 4913 < N$  — no wraparound, so  $C$  is the exact integer  $m^3$ .  $\sqrt[3]{4913} = 17 = m$ . **No private key touched.**

**Practical consequence (one line for the board).** Never send the same/related message to multiple recipients under small  $e$  without randomized padding (OAEP fixes this — Lesson 3).

---

## 2. Common Modulus Attack

(reuses: Extended Euclidean Algorithm)

**Setup.** One shared modulus  $n$ , two users with coprime exponents  $e_1, e_2$  ( $\gcd(e_1, e_2) = 1$ ), same message encrypted under both:

$$c_1 = m^{e_1} \bmod n, \quad c_2 = m^{e_2} \bmod n.$$

**Core insight (say this first).**  $\gcd(e_1, e_2) = 1$  means Extended Euclid can write  $1 = ae_1 + be_2$  — the *exact same computation* used to build the private key  $d$ . Here it's aimed at the exponents directly: combining  $c_1, c_2$  with exponents  $a, b$  combines  $m^{e_1}, m^{e_2}$  into  $m^1 = m$ .

**Theorem 2.** If  $\gcd(e_1, e_2) = 1$ , anyone knowing only  $(n, e_1, e_2, c_1, c_2)$  recovers  $m$  — no private key at all.

**Compressed proof.**

- Extended Euclid on  $(e_1, e_2)$  gives integers  $a, b$  with  $ae_1 + be_2 = 1$ .
- $c_1^a c_2^b \equiv (m^{e_1})^a (m^{e_2})^b \equiv m^{ae_1 + be_2} \equiv m^1 \equiv m \pmod{n}$  — pure exponent-law algebra, no secret used.
- If (say)  $b < 0$ : compute  $c_2^{-1} \bmod n$  first (Extended Euclid again), then raise *that* to  $|b|$ .

**Worked example (verified — corrected from earlier draft).**  $n = 3233$  ( $p = 61, q = 53$ ),  $e_1 = 17, e_2 = 7, m = 42$ :

$$c_1 = 42^{17} \bmod 3233 = 2557, \quad c_2 = 42^7 \bmod 3233 = 240.$$

Extended Euclid(17, 7):  $a = -2, b = 5$  (check:  $-2 \cdot 17 + 5 \cdot 7 = 1$ ). Since  $a < 0$ :  $c_1^{-1} \bmod n = 3123$ . Then

$$m = (c_1^{-1})^2 \cdot c_2^5 \bmod n = 42. \checkmark$$

(Note: the lecture PDF previously had a typo here —  $c_1^{-1} = 2160$  — now fixed to 3123 in your source file; 3123 is the value that actually makes the arithmetic check out.)

**Practical consequence.** Never share a modulus across users/purposes, even with different exponents — broken the instant any two exponents on it are coprime (the common case).

---

## 3. Cycling Attack (Conceptual)

(reuses: Euler's theorem)

**Setup.** Only  $(n, e, c)$  with  $c = m^e \bmod n$ . Repeatedly re-encrypt:  $c_1 = c^e \bmod n, c_2 = c_1^e \bmod n, \dots$  until the sequence returns to  $c$ .

**Core insight (say this first).** This *is* Euler's theorem, one level up. An hour ago: repeatedly multiplying by  $a$  cycles back to 1 after  $\varphi(n)$  steps. Now: repeatedly *encrypting* (exponentiating by  $e$ ) must also cycle back eventually, since encryption is invertible. The punchline: the plaintext sits in plain sight, one step before the cycle closes.

**Theorem 3.** Let  $k$  be smallest with  $c_k = c$ . Then  $m = c_{k-1}$ .

**Compressed proof.**

- Induction:  $c_i \equiv m^{e^{i+1}} \pmod{n}$ .
- Definition of  $k$ :  $m^{e^{k+1}} \equiv m^e \pmod{n}$ , i.e.  $e^{k+1} \equiv e \pmod{\text{ord}_n(m)}$ .

- $e$  is coprime to  $\varphi(n)$  (chosen that way at keygen), hence coprime to  $\text{ord}_n(m) \mid \varphi(n)$  too  $\Rightarrow$  cancel one factor of  $e$ :  $e^k \equiv 1 \pmod{\text{ord}_n(m)}$ .
- So  $c_{k-1} \equiv m^{e^k} \equiv m^1 \equiv m \pmod{n}$  — “one full lap plus one step back,” exactly the Euler’s-theorem clock picture from 1.1.

**Worked example (verified).**  $p = 47, q = 59 \Rightarrow n = 2773, \varphi(n) = 2668, e = 17, m = 123$ :

$$c = 123^{17} \bmod 2773 = 1924.$$

Repeatedly raising to the 17th power: after **44** iterations the sequence returns to  $c_{44} = 1924 = c$ . One step earlier,  $c_{43} = 123 = m$ . **Recovered.**

**Why it stays conceptual (say this explicitly, don’t let it undersell the point).** At real key sizes the cycle length is astronomically large — utterly infeasible in practice. It’s taught because it’s the cleanest possible demonstration that RSA encryption is literally iteration inside a finite group, and it previews why  $\lambda(n)$  having a large prime factor matters (same spirit as avoiding smooth  $p - 1$ , Lesson 1).

---

**One-slide recap, if you want to close the loop:** all three attacks above use tools from *today’s earlier hour* (CRT, Extended Euclid, Euler’s theorem) as *attack* tools rather than *construction* tools — the exact sentence to leave students with before next time, when Wiener’s attack (continued fractions) opens properly instead of being rushed in.